AI Tool Prompt Engineering Report

21st Century IT Skills

ELECTIVE 3 - GE 4120

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Submitted to:

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April 2026

AI TOOL OVERVIEW

In an era where athletic performance is inextricably linked to technological innovation, the modern bicycle has evolved from a simple mechanical machine into a marvel of fluid dynamics. I engaged an AI, specifically Perplexity AI, to explore this evolution, specifically the profound influence of aerodynamics on the cycling industry because synthesizing the complex blend of physics, engineering, and sports history requires a tool capable of processing vast amounts of technical data. I chose Perplexity AI for this research because its ability to quickly synthesize dense technical data and provide precise, citation-backed answers makes it the ideal tool for exploring the complex engineering behind modern cycling.

Original Prompt: how can aerodynamics influence current time trial bicycles?

Result:

how can aerodynamics influence current time trial bicycles?

Aerodynamics profoundly shapes modern time trial bicycles by minimizing air resistance, which accounts for up to 90% of a rider's power expenditure at racing speeds above 40 km/h. Key design choices prioritize drag reduction over other factors like weight or comfort. bikeradar +1

Frame Design

Frames use Kammtail tube profiles—truncated airfoils that mimic longer shapes for better airflow—enabling deeper sections without UCI rule violations. Steeper seat tube angles (75-78 degrees) position riders lower and more forward to shrink the frontal area. youtube bikeradar bikeradar

This aerodynamic TT bike showcases deep wheels and integrated cables that further streamline airflow.

Wheel Optimization

Deep-section rims and disc rear wheels cut drag significantly; rotating wheels interact with the frame to reduce front wheel drag by up to 12.9% compared to static tests. Aero wheelsets alone can save 1-2 minutes over a 40 km TT versus standard wheels. researchportal.hw +1

Rider Integration

Bike geometry complements aggressive positions with clip-on aero bars, where even small posture tweaks—like reduced hip angle—yield net gains despite higher physiological cost. Rider position changes can improve 40 km times by over 6 minutes by slashing drag area (CdA). pubmed.ncbi.nlm.nih +2

AI-ASSISTED PROMPT

I utilized Google's Gemini to refine my initial questions. By inputting my original, somewhat broad prompt into the LLM and asking for its feedback, Gemini helped me clarify my intent, add necessary technical context, and structure the phrasing more effectively. This iterative process transformed my basic inquiry into a highly optimized and specific prompt, ultimately allowing me to extract a much richer and more focused analysis from my research tools.

Original Prompt: How can aerodynamics influence current time trial bicycles?

Improved Prompt: How do aerodynamic principles influence the design, component integration, and real-world performance of modern time trial bicycles, and what are the key trade-offs involved?

Result:

how can aerodynamics influence current time trial bicycles?

could you improve my prompt?

Here's a sharper, more effective version of your prompt, depending on what you want to get out of it:

🖊️ **Improved general version**

"How do aerodynamic principles influence the design and performance of modern time trial bicycles?"

🎯 **If you want a deeper, more analytical answer**

"How do aerodynamic principles influence the design, component integration, and real-world performance of modern time trial bicycles, and what are the key trade-offs involved?"

🧠 **If you want something more technical/engineering-focused**

"In what ways do aerodynamic principles—such as drag reduction, airflow management, and boundary layer control—shape the design and efficiency of modern time trial bicycles?"

🚴 **If you want it more practical/applications-based**

"How does aerodynamics affect the speed and efficiency of modern time trial bicycles, and which design features contribute most to performance gains?"

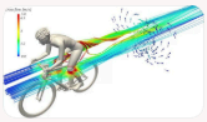
If you tell me your context (essay, research paper, presentation, etc.), I can refine it even further or tailor it to a specific word count or level.

How do aerodynamic principles influence the design, component integration, and real-world performance of modern time trial bicycles, and what are the key trade-offs involved?

Aerodynamics dominates modern time trial (TT) bicycle design by slashing drag, which comprises 80-90% of resistance at racing speeds over 40 km/h. Principles like minimizing frontal area, smoothing airflow via Kammtail tubes, and integrating rider position drive innovations, though trade-offs limit universality. bikeradar +1

Design Principles

Frames adopt truncated airfoil (Kammtail) shapes for deep sections within UCI rules, reducing pressure drag while maintaining stiffness. Geometry features steep seat angles (75-78°) and low stacks to optimize rider posture, cutting CdA (drag coefficient times area) by up to 20-30% versus road bikes. minds.wisconsin +3 chainreactio

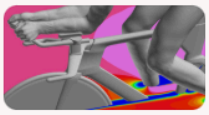


Bikepacking bicycle airflow simulation chainreactionbicycl

CFD simulations like this reveal airflow streamlines around rider-bike systems, guiding tube shaping to accelerate air and reduce wake turbulence.

Component Integration

Wheels use deep rims (50-80mm front, disc rear) for 10-15% less drag, with rotation effects amplifying savings in real yaw conditions. Cables route internally, bars integrate hydration, and tires balance low rolling resistance with aero width (25-28mm); rider position contributes 70-80% of total drag, customized via 3D fitting. csbikes +2 argon18



Integrated aero bars and frame channels in this render minimize separation bubbles for seamless flow.

Performance Gains

Aero setups save 1-3 watts per component, yielding 1-2 minutes over 40km TTs; full optimizations (bike + position) can cut times by 5+ minutes. Wind tunnel and CFD validate gains at yaw angles (5-15° crosswinds). minds.wisconsin +2

Key Trade-offs

Aspect	Aero Benefit	Trade-off
Weight	Minimal (deeper tubes add 200-500g)	Slower climbs; irrelevant on flats chainreactionbicycles +1
Comfort/Stability	Streamlined posture reduces CdA	Higher physiological cost, less handling on technical courses tecquipment +1
Cost/Complexity	50-100W total savings potential	$10k+ bikes; custom fitting required csbikes
Versatility	TT dominance (95% drag focus)	Unsafe/ineffective for group rides or crosswinds sevencycles

Real-world data questions peloton aero gains due to yaw/drafting, favoring TT purity. chainreactionbicycles +1

13 sources

COMPARISON AND ANALYSIS

The comparison between my original prompt and the improved AI-assisted prompt highlights the significant impact that effective prompt engineering can have on the quality and depth of generated responses. Initially, my prompt “How can aerodynamics influence current time trial bicycles?” was straightforward but somewhat broad and limited in scope. While it was able to generate a general explanation, the response lacked depth, specificity, and a comprehensive exploration of the topic. It primarily focused on basic aerodynamic concepts without fully addressing the complexity of modern bicycle design.

Through the assistance of Google’s Gemini, my prompt was refined into a more detailed and structured question: “How do aerodynamic principles influence the design, component integration, and real-world performance of modern time trial bicycles, and what are the key trade-offs involved?” This improved version demonstrates a clearer intent and introduces multiple dimensions of analysis. By specifying areas such as design, component integration, performance, and trade-offs, the prompt guided the AI to produce a more thorough and insightful response.

The difference in output between the two prompts was substantial. The improved prompt resulted in a more organized and in-depth explanation, covering not only the theoretical aspects of aerodynamics but also its practical applications in real-world cycling scenarios. It also addressed important considerations such as the balance between speed, comfort, and control, elements that were not emphasized in the original response. This shows that a well-crafted prompt can significantly enhance the relevance and richness of the information provided by AI tools.

Overall, this exercise demonstrates that prompt engineering is a crucial skill when working with AI systems. A more precise and thoughtfully constructed prompt enables the AI to better understand the user’s intent, resulting in more accurate and meaningful outputs. This experience has shown me the importance of refining questions and being specific in order to fully leverage the capabilities of AI in research and analysis.